

2(a)(i)	1 is regulatory gene / codes for repressor (protein) ; R is a transcription factor R <i>lacI</i> is a repressor protein 2 prevents, transcription / RNA polymerase binding to promoter ;	2
2(a)(ii)	any three from: <i>concentration of IFN-α produced</i> 1 increases steeply (after addition of IPTG) ; 2 peak is, at 8 hours / 4 hours (after addition of IPTG) ; 3 decrease is less steep or final concentration greater than starting concentration / AW ; 4 data quote ;	3
2(a)(iii)	any one from: <i>IPTG is higher / lactose is lower, because</i> 1 lactose has to be converted to allolactose ; 2 lactose is broken down so needs to be continually taken up or IPTG is not broken down so continually binds to repressor ; 3 IPTG will be at higher concentration than allolactose ; A lactose for allolactose 4 IPTG has higher affinity for repressor protein than allolactose ; A lactose for allolactose 5 lactose, used up / concentration decreases ; 6 more IPTG enters ;	1

Question	Answer	Marks
2(a)(iv)	IFN- α is, unstable / breaks down ;	1
2(b)	(only <i>E. coli</i> that have taken up the plasmid) will, grow / survive, in the presence of, ampicillin / antibiotic ;	1
2(c)	any five from: 1 (random) mutation ; 2 natural selection / directional selection ; 3 antibiotic acts as selection pressure / AW ; 4 bacteria with mutation, have selective advantage / survive / reproduce or bacteria with, gene / allele, that codes for antibiotic resistance, have selective advantage / survive / reproduce ; 5 pass on, mutation / gene / allele, (for antibiotic resistance) by, binary fission / asexual reproduction / vertical transmission ; 6 pass on, mutation / gene / allele, (for antibiotic resistance) by, transduction / transformation / conjugation / horizontal transmission ; 7 AVP ; e.g. increased chance of resistance if people do not finish full course of antibiotics overuse of antibiotics some antibiotics may act as mutagens	5